

Spectrograms as Tokens

DATA 443/643 · Week 6, note 6.1

Week 6

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Ideas from Stanford CS224S 2025 L2 (spectrogram as spectrum + time) and L5 (why not a raw waveform). Original slides; that course is not copied.

Lecture plan

- 1 A 1-D wave, then an STFT
- 2 Count frames, not samples
- 3 Patch the picture as in ViT
- 4 Whisper's 25 ms / 10 ms numbers

Key goal: leave able to compute T from L, N, H, and to refuse a 16 kHz transformer.

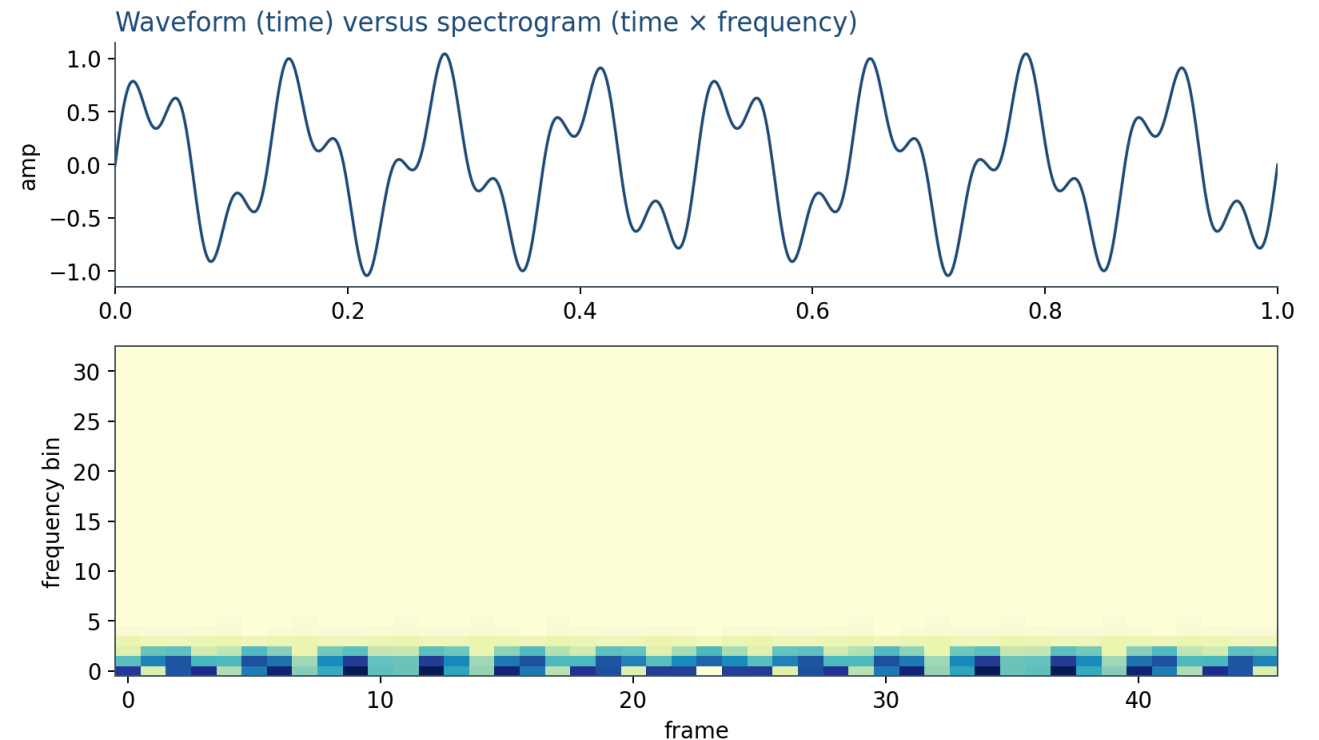
First time: what / why / formula

- What / why. STFT: window the waveform, FFT each frame, plot frequency vs time. A mel spectrogram warps frequency to a perceptual scale. Waveforms are long and oscillatory. A 2-D time–frequency map can be patched like a ViT image.
- Architecture. Waveform $\rightarrow$ frames $\rightarrow$ STFT magnitudes (optionally mel + log) $\rightarrow$ patch embed $\rightarrow$ transformer.
- Formula. Frame m : $X(m, \omega) = \sum_n x[n]w[n - mH]e^{-j\omega n}$. Sequence length $\approx$ number of frames (or patches of frames).
- Tradeoffs. + Shares ViT code. – Phase often dropped; time resolution vs frequency resolution; not yet a transcriber (next note).

How. Lab 6: sine mixture, STFT, patch. Hop length and window are method knobs.

A wave, then a picture

- Pressure $x[n]$ at sampling rate f_s .
- STFT: window length N , hop H , then a DFT.
- Plot magnitudes. Most encoders drop phase.



How many frames?

$$T = 1 + \text{floor} \left((L - N) / H \right)$$

One second at 16 kHz:
 $L = 16000$.

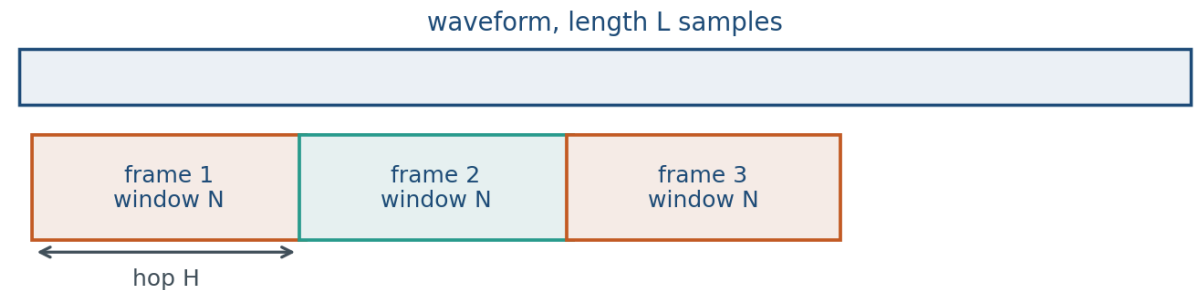
$N = 400, H = 160$ gives
 $T = 98$.

Double the hop: about half as many frames. Frequency bins do not move.

Slide the window

- Window $N / f_s = 25$ ms, hop 10 ms: Whisper's default.
- Frames last long enough to see a phoneme.
- A raw 16 kHz wave is 16,000 tokens per second.

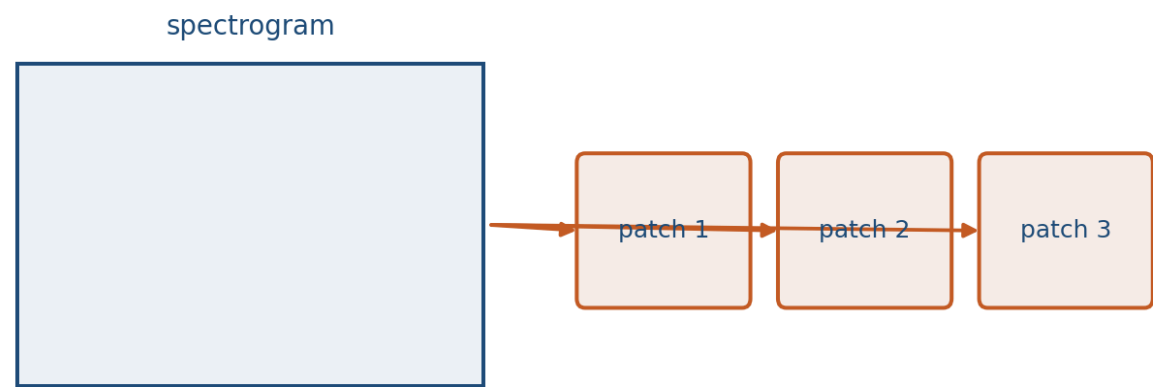
STFT: slide a window of length N by hop H . Frames $T = 1 + \lfloor (L - N)/H \rfloor$.



Then patch, as in ViT

- An 80×400 mel grid, 16×16 tiles: 125 tokens.
- Time is not quite space. Name which axis is frequency.
- Lab 6 crops remainders. Do not drop a formant band silently.

Same trick as ViT: cut time–frequency patches, then embed



What to take from this lecture

Write the front end

f_s , N , hop, mel versus linear.

Bin k

Near $k \cdot f_s / N$ hertz. Check Lab 6's 440 and 880.

Not a transcript

Tasks sit on top of the tokens. That is 6.2.